Data Analysis of Global Land and Ocean Temperatures From The Years 1750 to 2015

CPS188 - Computer Programming Fundamentals

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Introduction

In this report, the temperature of both the land and the ocean over the course of several years will be analyzed. This will be done by taking different averages based on different time periods, whether that be by century, year, or month. Doing so, specific trends can be followed and conclusions can be drawn on whether or not average temperatures have risen or dropped as time goes on. This information is very useful for many audiences, whether that be a professional studying climatology, or a concerned member of the public who wishes to understand the effects of climate change. With this information provided in the report, people can further understand what kind of climatic trends are taking place in different time periods, and whether or not they are affected by them.

Preparation

Before approaching this problem, a few steps were taken in place in order to store information for easier reading.To put it simply, the GlobalTemperatures.csv file was read through until the end of the file and the delimited information was stored in 2D arrays to access information in a efficient manner. Arrays such as, store(to store integers) and first section (to store the dates) were accessed where the first dimension was used for the rows and the second dimension was used for the columns. To access different columns, a number was defined based on the position of the excel file. For instance, LAVGT represented the land average temperature column in the excel file. Arrays such as year extract and month extract were used to store all years and months as an integer for easier reference. For the GNU plots to read the file, the data was put into a txt file for clear reading.

Question 1 (Jenna)

Operation:

To obtain the yearly averages for the land average temperature column for the years between 1760 and 2015 nested for loops were used to calculate the averages in the years. The first nested for loop keeps track of the years between 1750 and 2015 while the second for loop runs through the whole file. Within the second for loop if the years in the file had matched the years in the first for loop it would store the sum of all the temperatures from that year and increase a counter. Then the average temperature for that year was calculated and printed out.The for loop would reset the variables and repeat these processes until all the yearly average temperatures were printed out for the years 1760 and 2015.

Simplified Process:

- Create nested for loops. One for the range of the years and one to run to the end of the csv file.
- If the years matched the beginning of the for loop sum all the temperatures for that year.
- Calculate the average for that year and print out the results.
- Reset variables and repeat the process.

Output: (Condensed due to the length of the output)

```
Question 1
Years:          Land Average Temperature for each year:
1760            7.185
1761            8.772
1762            8.607
1763            7.497
1764            8.400
1765            8.252
1766            8.406
1767            8.222
1768            6.781
1769            7.695
1770            7.692
1771            7.853
1772            8.194
1773            8.222
1774            8.772
1775            9.183
1776            8.304
1777            8.256
1778            8.542
1779            8.983
1780            9.433
```

By observing the results from question 1 it is clear that as the years pass by the temperature is steadily increasing. Nearing the 2000's it is shown that the temperature has increased to 9 degrees and achieving higher temperatures as the years go on. This data is crucial to analyze to observe for the extrapolation of future yearly temperatures to combat issues such as climate change.

Question 2 (Gayarube)

Operation:

The average of a set of data is obtained from dividing the sum by the number of data points. A sum array of double variable type and dimensions of 4 was created to store the sum of the temperatures of each century. The dimension was 4 as there are 4 centuries and the index indicates the century. For instance, index 0 represents the 18th century as arrays are zero-indexed, meaning that they begin at 0. A double variable type was used because the data is in decimal form. Similarly, a count array of integer variable type and dimension of 4 was created to find the number of temperatures for each century. The sum and count array are initialized at 0 and updated in the if statements. A for loop is used to iterate through all the temperatures in the LandAverageTemperature column of the excel sheet. The i represents the index for extracting the years of the store array. Inside the for loop, if statements are used to separate the different centuries. If statements are used to check if the current year is between the appropriate years of the century. If the statement is true, then it proceeded to the following operations inside the if statement. Inside the if statement, the temperature of the current year was added to the sum array and the count array was incremented by 1. After the for loop ended, the averages were calculated by dividing the sum array by the count array. The averages of each century were then printed to the output.

Simplified Process:

- Initialized and declared the sum and count variables
- Used a for loop to go through all temperatures. If statements with conditions were used inside the for loop to differentiate between each century. Continuously update the sum and count variables.
- Calculate the average and print to the output

Output:

```
Question 2
Average Land Temperature for the different centuries
The average land temperature for the 18th century (1760-1799) is 8.215.
The average land temperature for the 19th century (1800-1899) is 8.009.
The average land temperature for the 20th century (1900-1999) is 8.638.
The average land temperature for the 21st century (2000-2015) is 9.542.
```

The average land temperature appears to be slightly increasing, however there is a decrease between the 18th century and 19th century. The decrease may be due to the fact that the 18th century collected less temperatures than the 19th century, thus the averages are calculated on different scales. The averages are relatively similar and the increasing pattern makes sense as the temperature increases with global warming and the human contribution to the mass production of materials and resources.

<u>Question 3 (Jenna)</u>

Operation:

To obtain the land average temperatures monthly averages for the years 1900 and 2015, nested for loops were used to to keep track of the averages for each month. One for loop keeps track of the range of years while the other inner for loop runs until the end of the file.Inside the inner for loop,if the years at a certain index had matched the outer loops year; it would check for which month to add the temperature based on a series of if statements checking each month. Once all the temperatures for each month were totaled the averages of each month were printed out.

Simplified Process:

- Create nested for loops. One for the range of the years and one to run to the end of the csv file.
- If the years matched the beginning of the for loop,then check the month that corresponds based on the row's index number.Add the land average temperature for that month
- Calculate the average for each of the months then print out to the user.

Output:

```
Question 3
The averages for the land average tempurature from 1900-2015:
JANUARY: 2.82
FEBURARY: 3.33
MARCH: 5.40
APRIL: 8.54
MAY: 11.39
JUNE: 13.54
JULY: 14.45
AUGUST: 13.96
SEPTEMBER: 12.17
OCTOBER: 9.53
NOVEMBER: 6.22
DECEMBER: 3.81
```

By observing the monthly average temperatures for the years 1900-2015 it is shown that the temperatures have a higher temperature than from other years.This may be due to the fact that the years taken were from 1900-2015 which had the peak times for sudden industrialization and production of greenhouse gas emissions. For a better analysis, looking at different monthly temperatures for certain centuries can have a better pattern recognition of why temperatures increase. (I.e. moments in history).

Question 4 (Gayarube)

Operation:

Multiple arrays were initialized and declared to 0, and later used to be compared and updated such as the hottest and coldest months. A month array with a char variable type is initialized and declared with all twelve months, to easily convert the month from a number form to a word form. The for loop iterates through all the temperatures in the LandAverageTemperature column of the excel, stopping until it reaches the final temperature, and would follow the if statements within. The first if statement checked if the current temperature is greater than the highest temperature initialized previously. If the condition was true, the current temperature, month and year would replace the hottest temperature, month and year. The second if statement followed the same procedure however would compare and exchange values with temperatures less than the lowest temperature. When the hottest and coldest month and year are printed to the output, the month array is called and the index is the hottest/coldest month subtracted by 1 because the array begins at index of one.

Simplified Process:

- Arrays initialized and set to 0 to be compared with the current values in the if statements.
- For loop created with if statements, comparing the current temperature with the hottest or coldest temperature. Values would be replaced if conditional statements were true.
- Print the coldest and hottest months and years.

Output:

```
Question 4
Determine the hottest and coldest month recorded. Mention the month and year. If a tie, mention only one.
The hottest month is July during the year 1761.
The coldest month is January during the year 1768.
```

From the results recorded, the hottest month and coldest month agree with the seasonal temperatures experienced during current time. The hottest months tend to be July and August which are the summer months and winter occurs from the end of November to the end of January, which satisfy the results obtained. Since the first hottest and coldest month was required, these years appear to be in the 18th century according to the data format in the file.

<u>Question 5 (Lianne)</u>

Operation:

To obtain the highest and lowest temperatures from the years 1760-2015 loops and certain array's were used to iterate through years to find the highest and lowest temperatures. The average was used in the first question to store the yearly average temperatures from 1760-2015.Through the use of for loops to keep track of the range of years the if statement checks if the temperature the year is the highest or lowest; if true make set the year for it to be the maximum or the minimum (where maximum corresponds to the hottest year and minimum refers to the lowest year).After iterating through the years the result is printed to the user.

Simplified Process:

- Set for loop for the range of the years
- Check if the average temperature for the year that is iterated is the highest or lowest temperature.
- If true, make the maximum index to the iterated year for the hottest temperature in the average array.If it is the coldest temperature make it the minindex.
- Print out the result to the user.

Output:

```
Question 5
The highest tempurature is 9.831000 corresponding with the year 2015
The lowest tempurature is 6.781333 corresponding with the year 1768
```

By observing this data it is shown that the year that has the highest temperature is 2015 and the year with the lowest temperature is 1768.This is to be expected due to the advancement of industrialization and the production of greenhouse gasses.Through observing this data, the years have a drastic increase of around 3 degrees which is concerning when analyzing climate change over the years. This shows that if there is no change or action towards climate change that temperature will increase drastically.

<u>Question 6 (Lianne)</u>

Operation:

In order to evaluate the yearly average temperatures, some calculations had to be done in code. Essentially, using for loops each year was split into 12 months, and the respective temperature columns were put into a function that calculates the yearly average temperature given the data from 12 months in a year. This process was done for the land average temperature column, land max temperature column, land min temperature column, and the land and ocean average column. This had to be done for more than just the average land temperature, as evaluations can be done on both the highest and lowest land temperatures recorded, as well as the land and ocean temperatures. In the end, a txt file, titled 'YearlyAverages.txt' was created that stored all the different yearly temperature averages from years 1760-2015. The purpose of this txt file is so that the information can be graphically portrayed in GNUplot.

Simplified Process:

- Using for loops and if statements, find the respective temperatures for the 12 months in a year.
- Once the sum of those temperatures is received, it is placed into an equation to find the yearly average for the respective temperature columns.

- In an organized manner, print the results into a new txt file, labeled ‘YearlyAverages.txt’, where the year is listed in one column, and the temperatures are listed in their own respective columns.

Output:

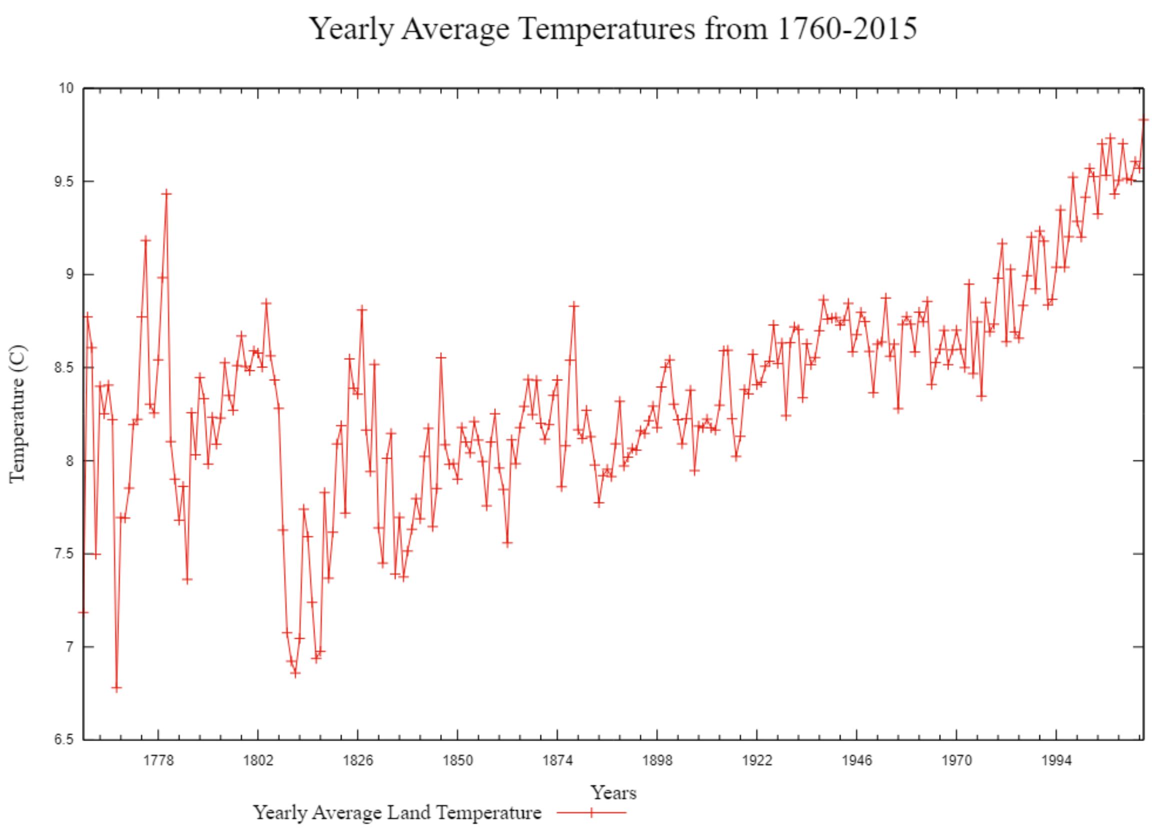

Figure 1 - A graph showing the yearly average land temperatures from 1760-2015.

GNUPlot was used to display the yearly average temperatures from the year 1760 to 2015. The GNUPlot script for the labels involves setting the size of the picture using svg printing the image directly on the website, using the term “key” to show what each line represents, and adding axis labels using “xlabel” and “ylabel”. To plot the actual data, “xdata time” takes in the information from the year column of the txt file, and executes using placeholder “%Y”. Other formatting tools for plotting the data include “timefmt” and “format x” which helps display the values. The range is set using “xrange”, and to execute the graph the plot function is used as well as indicating the line point using “lp” and “lw” to determine the weight of the line.

Question 7 (Jenna)

Operation:

To make line plots of the 19th and 20th centuries, first a c code was established in order to calculate the averages for each of the centuries years.This was achieved by using nested for loops.The outer for loop focused on the bounds of the centuries while the inner for loop ran through the whole file. Inside the inner loop variables like tempstorer and counter2 were used to add land average temperatures according to certain years in between the range bounds.In the outer loop the average was calculated and stored in an array.As the outer loop repeats the variables clear to calculate for the next year.This was done for both the 19th and 20th century average values. From there a text file was open to write in the newly obtained values for the GNU code.

To establish the GNU code for both centuries in one plot the x-axis range was established from 1 to 99. This was to observe the years side by side, for instance the data between 1825 and 1925 can be easier to visually compare.The main title, x -labels, y-labels, and legend were established for easier legibility.The data points were plotted by making the reference to the newly created text file where column 1 holds the 19th century values and column 2 holds the 20th century values.

Simplified Process:

- Create nested for loops. One for the range of the years and one to run to the end of the csv file.

- If the years matched the outer for loop for the century ranges (for a specific year). Add the specific temperatures. And calculate the averages.
- Store each of the centuries averages in an array
- Print the century averages to a txt file for GNU plot to handle
- Plot the centuries on GNU plot using two different lines. Indicate using a legend.
- Add other graph parameters (title, legend, labels, ranges, and etc)

Output:

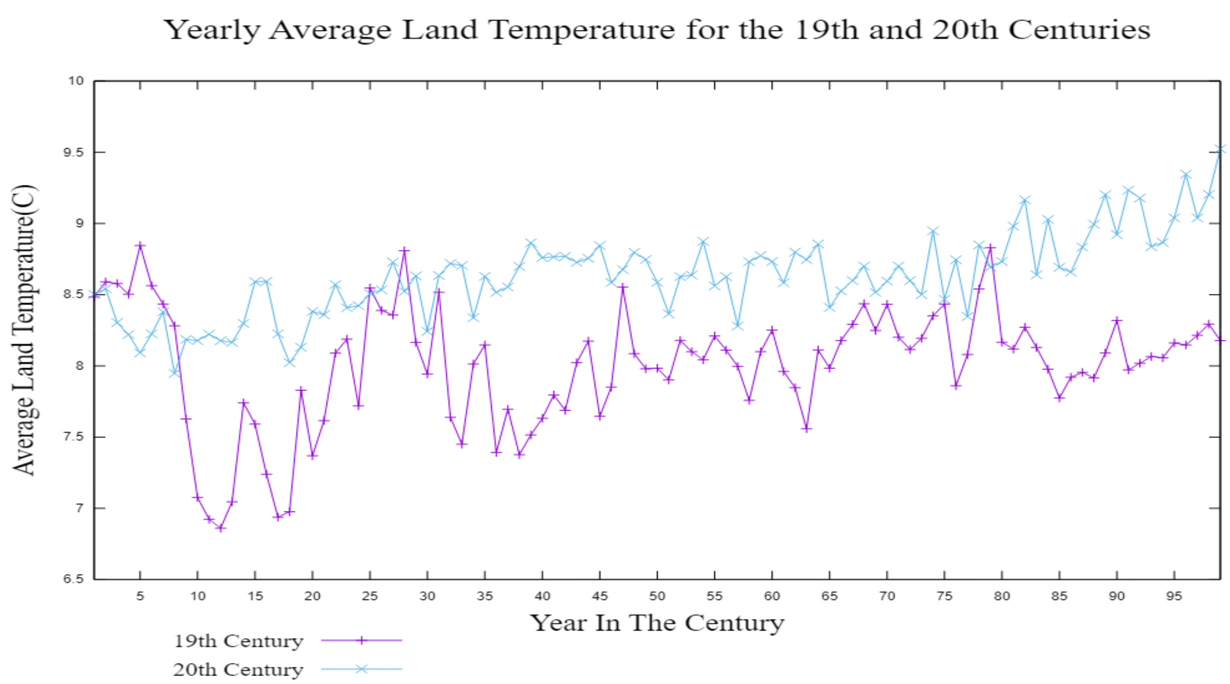

Figure 2 - Comparison of the 19th and 20th century in terms of their yearly average land temperature.

By observing the graph above it is shown that the 20th century has a steady increase in temperature as the years go by.For the 19th century the temperature seems to fluctuate. For instance, the 1810's had the lowest average land temperature while peak temperature had begun nearing 1806,1827, and 1877. To better understand why these temperatures fluctuate it is best to correlate certain data towards natural disasters, moments in history, etc.

Question 8 (Josip)

Operation:

Utilizing the “YearlyAverages.txt” file obtained from the C code in question 6, a graph was made using the columns “Average Land Temperature”, “Max Land Temperature”, and “Min Land Temperature”. Once again, the x-axis was formatted to contain only the year, just as it is in column 1 of the file. After plotting, to differentiate between the different temperature data, the function “lw” was used to make the yearly average land temperature more prominent than the other temperature data being plotted.

Simplified Process:

- Import “YearlyAverages.txt” into GNUplot
- Format the axis’ and label appropriately, and create a legend identifying what the plotted lines mean
- Using the line weight and line colour options, differentiate between the three lines, ensuring that the average land temperature is the most prominent out of the three.

Output:

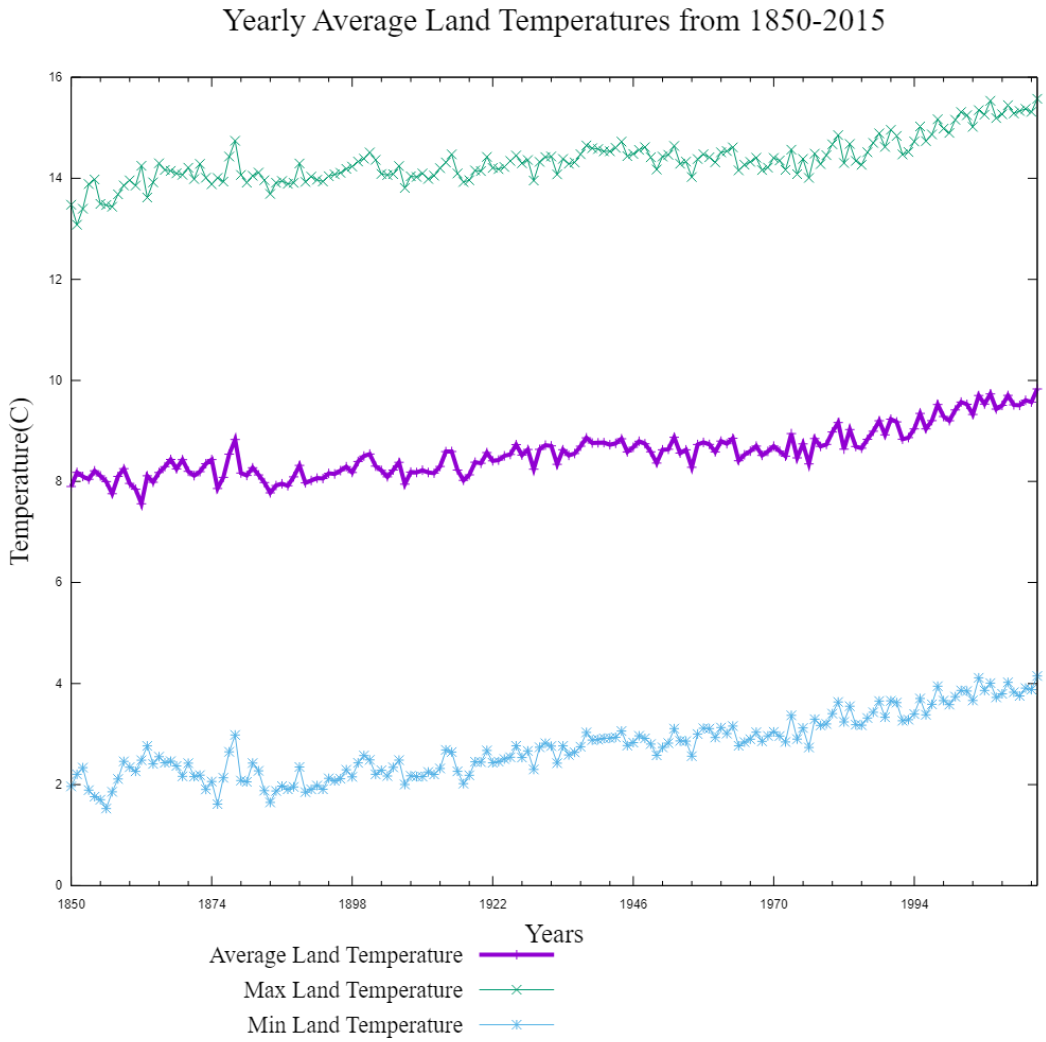

Figure 3 - Graph showcasing the average max temperature, minimum temperature, and average land temperature from the years 1850-2015.

In order to obtain the information from the graph above, the formatting of time in the first column in the txt file must be established in GNUplot with the command “set xdata time”, then “set timefmt “%Y”” to display years on the x-axis. Then, to obtain the specific time period, the xrange is set from 1850-2015. Finally, the respective columns are plotted and formatted to be different from one another by using tools like “lc” for line colour, and “lw” for line weight.

From figure 3, it can be seen that as time goes on, the average temperature rises. Especially from the time period of 1970-2015, it is seen that the average max temperature, which is essentially the highest temperature reached that year, is rising upwards to 16℃, compared to the start of the period being analyzed, which is around the temperature 13℃. The average land temperature also increased from approximately 8℃ to 10℃. Although this difference may seem minor, it is concerning that the temperature is rising as years pass.

<u>Question 9 (Gayarube)</u>

Operation:

To convert the C file to be used in GNU plot, a text file was opened. The average temperatures for each century were obtained in the same way as question 2, by determining the century and adjusting the sum and count variables to perform the calculation. Similar to question 4, the lowest and highest temperature for each century was obtained by using if statements within the if statements and else if statements, used to determine the century through conditional statements. The if statements compared the lowest and highest temperatures to the current temperatures in the LandMaxTemperature and LandMinTemperature columns. If the condition was true, the current temperature would replace the previously initialized one. The averages, low and high temperatures for the 19th, 20th and 21st century were printed to the text file and the text file was closed.

After the file was imported in the GNU plot, several properties were set in the GNU plot for the histograms as they shared similar characteristics including the size and filling. The x-axis and y-axis were accordingly labeled, where the century was labeled on the x-axis and the temperature was labeled on the y-axis. To combine all the subplots into one big plot, a multi layout had to be set. When plotting each histogram, an appropriate title was set along with an appropriate scale for the y-axis to enhance the data recorded. Specific columns of the text file were plotted, with a designated colour for each century and a legend recorded. The font of the text in the histogram was set to Times New Roman.

Simplified Process:

- Similar to previous questions 2 and 4, the average, low and high temperatures were obtained from comparing the current temperature with already declared values, which were iterated through a for loop.
- The file was converted from C to a text file, to be used in GNU plot.
- In the GNU plot, various characteristics were added to design each histogram and combined in a multi layout. To ensure each plot displayed the three centuries, the specific columns were called from the text file

Output:

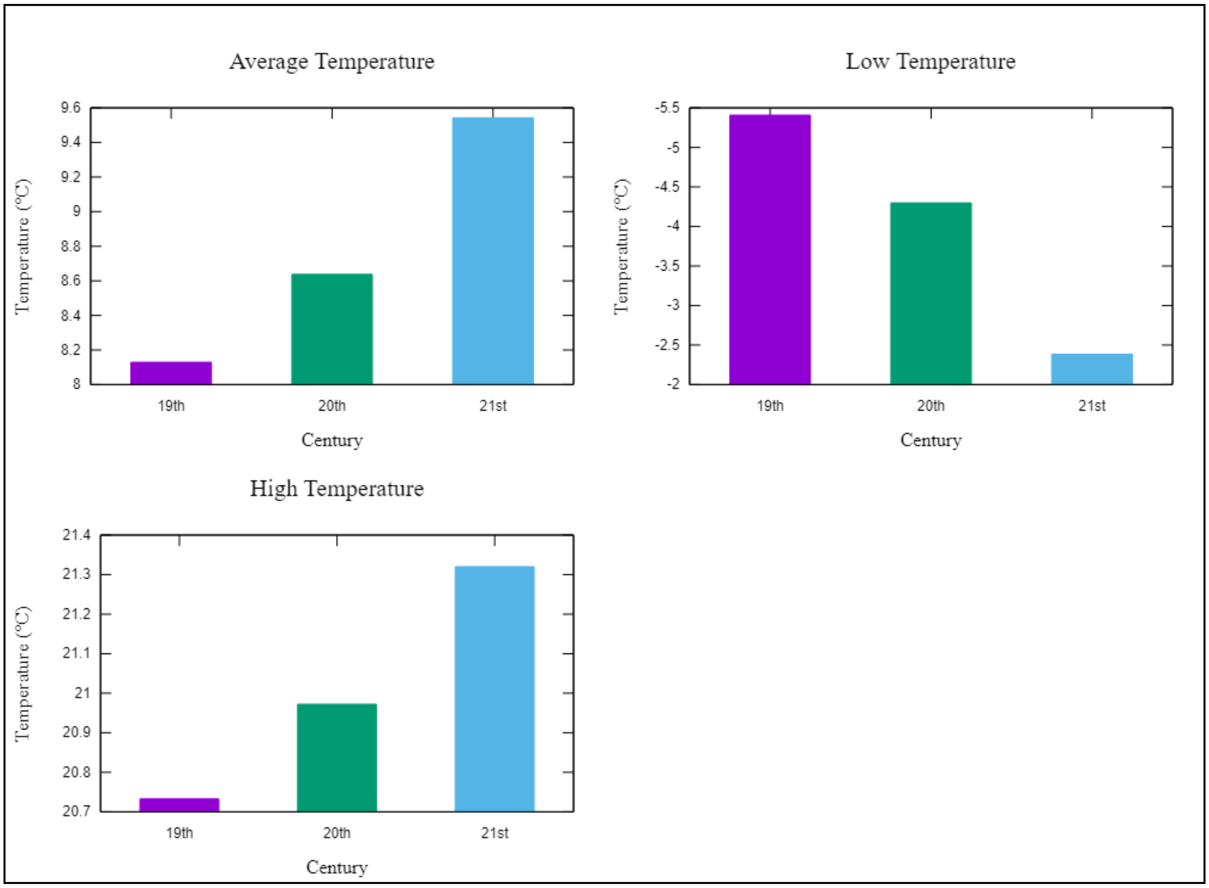

Figure 4 - The Average, Low and High Temperatures for each Century (1850-2015).

The incline in temperature can be observed as the centuries progress in the Average Temperature plot. The century in which appeared to have the highest temperature is the 21st century while the 19th century had the lowest temperature. The trends for the high temperature and low temperature graphs share an inverse relationship, where when the time increases (centuries), temperature increases simultaneously. This makes sense due the climate issues occurring in the world which fluctuate the temperature. For instance, global warming has a huge impact, as greenhouse gases are released into the atmosphere from burning fossil fuels, increasing the temperature of the planet.

Question 10 (Josip)

To begin, the txt file “GlobalTemperatures.txt” which was created by the conversion of the csv file was used. In this specific scenario, the plot required the x-axis to display the time as “YYYY-MM-DD”, which was done so by using “set xdata time”, then adjusting the time format using “set timefmt ‘%Y-%m-%d’” to match the first column of the “GlobalTemperatures.txt” file. Next, the actual plotting of data was done first with the error bars, then the actual temperature data. The uncertainty values were adjusted by a factor of 4 to allow them to be visible on the graph. Without doing so, the error bars appeared as if they were the points corresponding to the temperatures, rather than uncertainties.

Simplified Process:

- Format the axis time data, and set labels, titles, etc.

- Plot the error bars of the data being plotted, which were multiplied by a factor of 4 for visibility purposes.
- Plot the actual temperature vs. time data.

Output:

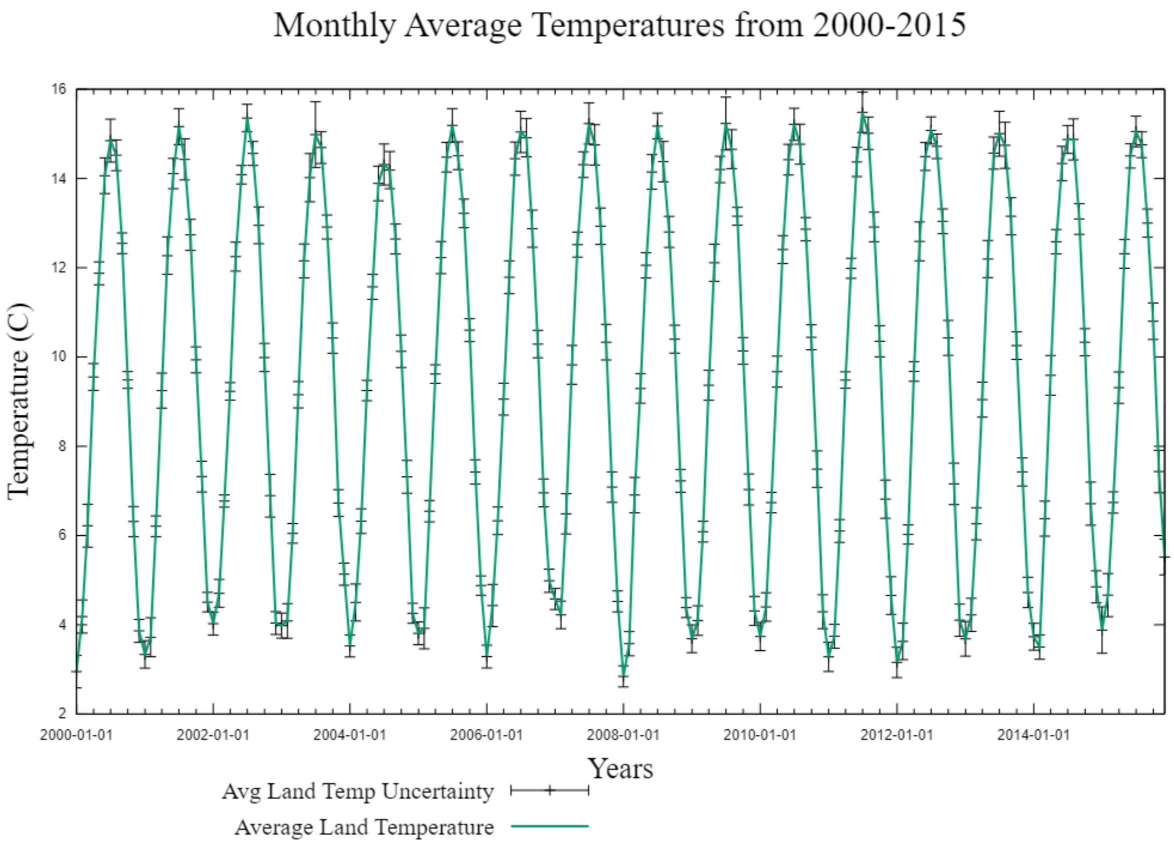

Figure 5 - Monthly average land temperature from the date 2000-01-01 to 2015-12-01, with uncertainty.

As the months go by in a year, the average monthly land temperature exhibits almost wave-like properties. In the colder beginning months, like January, February, and March, the temperatures are quite low, whereas it peaks around the summer months June, July, and August. Once summer is over, the temperature drops back down around October-November, and this pattern repeats. This trend has remained continuous throughout the 21st century, with only very slight temperature changes in the coldest and hottest months of the year.

Question 11 (Lianne)

Operation:

In this problem, GNUPlot was used to display the yearly average land and ocean temperatures from the year 1850 to 2015. The GNUPlot script for the labels involves setting the size of the picture using svg thus printing the image directly on the website, using the term “key” to show what each line represents, and adding titles using “xlabel” and “ylabel”. To plot the actual data, “xdata time” takes in the information from the year column of the txt file, and executes using placeholder %Y. Other formatting tools for plotting the data include “timefmt” and “format x” which helps display the values. The range is set using “xrange”, and to execute the graph the plot function is used as well as indicating the line point using “lp” and “lw” to determine the width of the line.

Simplified Process:

- Using the “YearlyAverages.txt” file, a new GNUplot file was made.
- Using formatting commands, the general outline of the graph, such as the axis’ and title, as well as legend were created
- The necessary columns were used to plot the corresponding information shown on figure

Output:

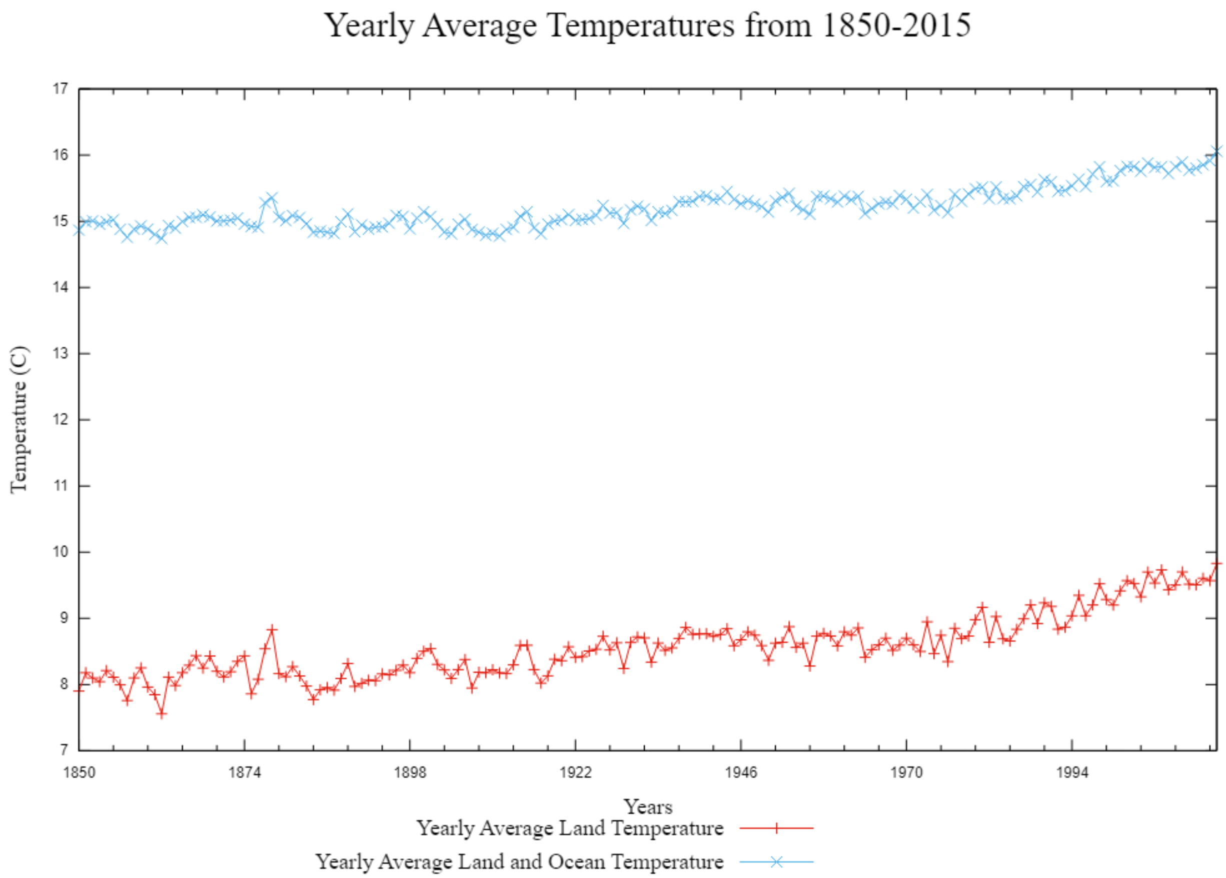

Figure 6 - A graphical representation of the yearly average land temperature, as well as the combined yearly average land and ocean temperature from 1850-2015.

From figure 5, it is evident that the average land temperature (red line) is much lower than the combination of the average land and ocean temperature (blue line). This means that the ocean temperatures are much greater than the land temperature. This might raise concerns in regards to aquatic ecosystems, which if harmed, also harms aquatic life.

Conclusion

From the data observed in the code, linear and bar plots, the pattern of the temperature can be observed for all the centuries. The average temperature appeared to be increasing as time increased, or as the centuries progressed. During the transition of the 18th century to the 19th century, there appeared to be a drastic drop in temperature according to the Yearly Average Temperatures from 1760-2015. This may have occurred due to a decline in sunlight and thus a decrease in solar radiation. The decrease of sunlight or heat energy created lower temperatures. Not to mention, the yearly average land temperature appeared to be significantly lower than the yearly average land and ocean temperature as time progressed. This can be observed in the Yearly Average Temperatures from 1850-2015. This can be due to the fact that oceans can store more heat than the land, thus there are less gradual fluctuations in the ocean temperature in comparison to the land. The main finding from the graphs plotted is the temperature gradually increased as time passed, which makes sense to the environmental issues the planet experiences due to climate change. As the temperatures of oceans rise and human activity continues to produce more greenhouse gasses to the atmosphere, the temperature will continue to rise due to the increase of waste produced.

To reflect on the experience of completing the project, a few steps can be taken to have led the project in a different direction. Scheduling more meetings to discuss the project and solve any minor issues could have helped progress the timeline at which the project was completed as well, eliminating any inconveniences. For instance, figuring how to convert the file from an excel sheet to a C file could have saved time. Working in a group project allowed the members to learn unique strategies in coding in C and how to effectively use resources to complete tasks. Not to mention, the teamwork skills gained from this project will aid the members in working in social environments and solving problems as a group. The project helped elevate transferable life skills that can help in any group setting in the future.